BACON BRO'S Design Doc for ***P04***

↳ Roster: Veronika Duvanova, Naomi Kurian, Alvin Sze, Andrew Tsai

PROJECT NAME: Lettuce Relax

TARGET SHIP DATE: 2026-4-20

Overview:

The initial load page will introduce you to food consumption and exercise worldwide through statistics, focusing on the health of all populations (debating whether or not to limit scope to youth or include everyone). The surprise is the interactive part where the visitor will be able to learn what food will provide them with what caloric intake, and what exercises they can do to burn off said calories (based on their traits?). The site will allow them to craft their own personalized dietary and workout goals, the progression of which will also be displayed with data visualization tools listed below.

Program Components

sqlite3 (Naomi):
user_info

Javascript (Alvin):
chart.js

Flask (Andrew):
__init__.py

HTML (Veronika + Andrew):
Log in + Sign up
Home
Food Nutrition data overview
Workouts data overview
Personalized projection

FEF (Veronika):
Bootstrap

Component Explanation:

sqlite3: used to store user info, including account information and tailored dietary/exercise plans.

chart.js is the javascript folder in which we will be able to create functions that take in parameters such as calories, nutrients, etc. and convert them into appropriate polar area, bar, and line graphs.

init.py is the main flask app handler, persistently hosted by nginx

Every page is dynamic because they have user-specific outcomes and/or require access to the ***user_info*** database to add/remove displayed information.

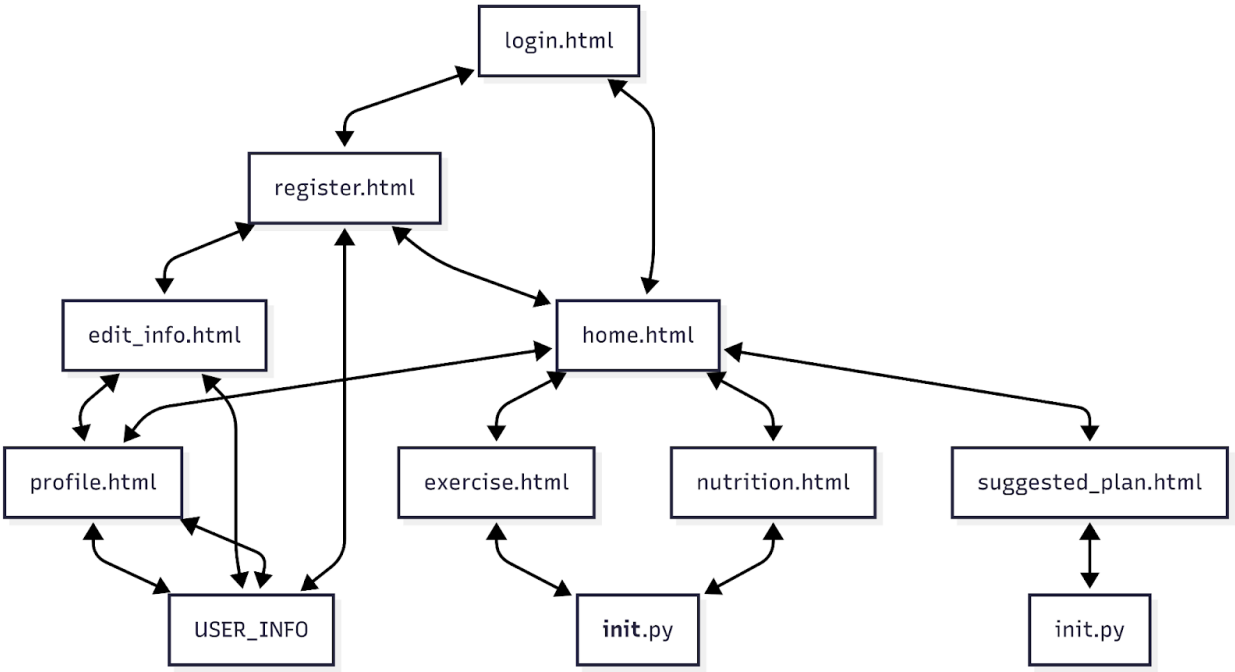
Datasets:

- FOOD NUTRITION: https://www.kaggle.com/datasets/utsavdey1410/food-nutrition-dataset (to provide data regarding foods and their nutrition, calories, etc)
- FAST FOOD:
- https://www.kaggle.com/datasets/ulrikthygepedersen/fastfood-nutrition (relates to fast food nutrition specifically, rather than all foods)
- WORKOUTS: https://www.kaggle.com/datasets/jockeroika/life-style-data (to provide information about workout times, instructions, and calories burnt)

Dataset Display Methods:

- **POLAR AREA:** https://www.chartjs.org/docs/latest/samples/other-charts/polar-area.html (to allow visual comparisons of individual foods' stats such as nutrients, calories, vitamins, etc)
- **BAR GRAPH:** https://www.chartjs.org/docs/latest/samples/bar/border-radius.html (to represent workout calorie burn)
- **Line Chart:** https://www.chartjs.org/docs/latest/samples/line/line.html (display projections set by personalized plan)

Component Map:



Database Organization

USER_INFO			
TEXT	user	PK	
TEXT	password		
TEXT	dietary_restrictions		
TEXT	exercises		

Based on dietary_restrictions and exercises, there will exist an algorithm that will create suitable workouts based on food intake or vice versa.

Site Map (front end):